AA 256 Design of Composite Structures

Homework #3

Due date: October 23rd, 2026

NOTE: Submit one PDF showing your methodology, calculations, and results. MATLAB or other calculation tools may be used.

1) Using the T800/3900-2 material, determine the A, B, D and α, β, δ matrices for the layups below. The following layups correspond to specimens in Project 1. (10 points)

a. [0/0/90/90]
b. [+30/−30/−30/+30]
c. $[45/-45]_s$
d. $[0/60]_s$
e. [0/45/−45/90]

The layups are listed from top to bottom; z is positive downward from the mid-plane. The subscript s denotes reflection about the mid-plane; stacks without s are listed in full.

Provide the general methodology for obtaining the A, B, D and α, β, δ matrices, and report the matrices with units. Results alone are not sufficient for full credit.

2) For a sufficiently long specimen, determine the tensile modulus (E_1) of each of the above laminates subjected to a tensile load P applied at +15° with respect to the 0° reference direction. Assume no applied moment and free bending. (10 points)

Material Properties	T800/3900-2
E_{xx} (msi)	23.2
$E_{\gamma\gamma}$ (msi)	1.30
$\nu_{x\gamma}$	0.28
$G_{x\gamma}$ (msi)	0.90
X_t (ksi)	412
X□ (ksi)	225
Y_t (ksi)	10
Y□ (ksi)	24.3
S (ksi)	10
h (in)	0.007

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E_x
E_y
ν_{xy}
G_{xy}

orthotropic, SO

$$Q = \begin{bmatrix} \frac{E_x}{1-\nu_{xy}\nu_{yx}} & \frac{\nu_{xy}E_y}{1-\nu_{xy}\nu_{yx}} & 0 \\ \frac{\nu_{yx}E_x}{1-\nu_{xy}\nu_{yx}} & \frac{\nu_y E_y}{1-\nu_{xy}\nu_{yx}} & 0 \\ 0 & 0 & G_{xy} \end{bmatrix}$$

z-axis rotation

$$T = \begin{bmatrix} \cos^2\theta & \sin^2\theta & 2\cos\theta\sin\theta \\ \sin^2\theta & \cos^2\theta & -2\cos\theta\sin\theta \\ -\sin\theta\cos\theta & \sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}$$

gives stiffness of each layer at its respective angle

$\bar{Q}_k = T_k^{-1} Q_k T_k$ for each layer k

obtaining A, B, D by integrating properties over all layers

$$A = \sum_k \bar{Q}_k (z_{k+1} - z_k) \quad [\text{Msi}\cdot\text{in}]$$

$$B = \sum_k \bar{Q}_k (z_{k+1}^2 - z_k^2) \quad [\text{Msi}\cdot\text{in}^2]$$

$$D = \sum_k \bar{Q}_k (z_{k+1}^3 - z_k^3) \quad [\text{Msi}\cdot\text{in}^3]$$

obtaining inverses

$$\begin{bmatrix} A & B \\ B & D \end{bmatrix}^{-1} = \begin{bmatrix} \alpha & \beta \\ \beta^T & \delta \end{bmatrix}$$

α: $\frac{1}{\text{Msi}}$ β: $\frac{1}{\text{Msi}\cdot\text{in}^2}$ δ: $\frac{1}{\text{Msi}\cdot\text{in}^3}$

results: see MATLAB attached

2) For a sufficiently long specimen, determine the tensile modulus (E_t) of each of the above laminates subjected to a tensile load P applied at +15° with respect to the 0° reference direction. Assume no applied moment and free bending. (10 points)

$\theta_{new} = \theta - \theta_{applied} = 0 - 15° = -15°$

since $e_1 = \alpha_1 N_1$ → $E_{11} = \frac{1}{\alpha_{11} H} = \frac{1}{\alpha_{11} h_n}$

with rotation of −15°

results: see MATLAB attached

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```
clc
clear
close all
format short g
```

Problem 1

```
Ex = 23.2; % Msi
Ey = 1.3; % Msi
G = 0.90; % Msi
nu_x = 0.28;

h = 0.007; % in

nu_y = Ey / Ex * nu_x;

Q = cast([
    Ex/(1-nu_x*nu_y)      nu_x*Ey/(1-nu_x*nu_y) 0;
    nu_y*Ex/(1-nu_x*nu_y) Ey/(1-nu_x*nu_y) 0;
    0 0 G
    ], "double");

T = @(x) [
    cos(x)*cos(x)   sin(x)*sin(x)  2*cos(x)*sin(x);
    sin(x)*sin(x)   cos(x)*cos(x) -2*cos(x)*sin(x);
    -sin(x)*cos(x) sin(x)*cos(x)   cos(x)*cos(x)-sin(x)*sin(x)];

angles = [
    0 0 90 90;
    30 -30 -30 30;
    45 -45 -45 45;
    0 60 60 0;
    0 45 -45 90
    ];

[N, N_layers] = size(angles);

[As,Bs,Ds,alphas,betas,deltas] = laminate_matrices(Q, angles, T, h);

names = {'A', 'B', 'D', 'alpha', 'beta', 'delta'};
units = {'Msi•in', 'Msi•in^2', 'Msi•in^3', '1/(Msi•in)', '1/(Msi•in^2)', '1/
(Msi•in^3)'};
matrices = {As, Bs, Ds, alphas, betas, deltas};
```

```
fprintf('\n==================== Problem 1 ====================\n');
for i = 1:N
    fprintf('\n==================== Specimen %d ====================\n', i);

    for j = 1:length(matrices)
        fprintf('\n%s [%s]:\n', names{j}, units{j});
        disp(matrices{j}(:,:,i));
    end
end
```

```
==================== Problem 1 ====================

==================== Specimen 1 ====================

A [Msi•in]:
      0.34451     0.010237   7.3713e-19
     0.010237      0.34451    1.812e-17
   7.3713e-19    1.812e-17       0.0252


B [Msi•in^2]:
   -0.0021557   3.3881e-21   5.1599e-21
   3.3881e-21    0.0021557   1.2684e-19
   5.1599e-21   1.2684e-19            0


D [Msi•in^3]:
   2.2508e-05   6.6882e-07   4.8159e-23
   6.6882e-07   2.2508e-05   1.1838e-21
   4.8159e-23   1.1838e-21   1.6464e-06


alpha [1/(Msi•in)]:
       7.2593      -0.2157   -1.996e-16
      -0.2157       7.2593   -1.714e-15
   -1.996e-16   -1.714e-15       39.683


beta [1/(Msi•in^2)]:
       695.24   3.6193e-16   -2.647e-14
  -2.5176e-15      -695.24  -5.8672e-14
   -2.647e-14  -5.8672e-14   1.7563e-28


delta [1/(Msi•in^3)]:
   1.1111e+05      -3301.6  -3.0551e-12
      -3301.6   1.1111e+05  -2.6235e-11
  -3.0551e-12  -2.6235e-11   6.0739e+05


==================== Specimen 2 ====================

A [Msi•in]:
```

```
      0.39204      0.11669            0
      0.11669     0.084083            0
            0            0      0.13165


B [Msi•in^2]:
            0            0            0
            0   2.7105e-20            0
            0            0            0


D [Msi•in^3]:
   2.5613e-05   7.6238e-06   9.5456e-06
   7.6238e-06   5.4934e-06   3.5224e-06
   9.5456e-06   3.5224e-06   8.6014e-06


alpha [1/(Msi•in)]:
       4.3461      -6.0315            0
      -6.0315       20.263            0
            0            0       7.5957


beta [1/(Msi•in^2)]:
  -1.2909e-14   5.2194e-14  -7.0486e-15
   4.3369e-14  -1.7535e-13   2.3681e-14
            0            0            0


delta [1/(Msi•in^3)]:
        86109       -78961       -63226
       -78961   3.1926e+05       -43115
       -63226       -43115   2.0408e+05


=================== Specimen 3 ====================

A [Msi•in]:
      0.20258      0.15218            0
      0.15218      0.20258            0
            0            0      0.16714


B [Msi•in^2]:
            0            0            0
            0            0   -5.421e-20
            0            0   -5.421e-20


D [Msi•in^3]:
   1.3235e-05   9.9421e-06   7.5448e-06
   9.9421e-06   1.3235e-05   7.5448e-06
   7.5448e-06   7.5448e-06    1.092e-05
```

```
alpha [1/(Msi•in)]:
        11.33      -8.5112   -2.491e-32
      -8.5112        11.33   3.3161e-32
   -2.491e-32   3.3161e-32       5.9831


beta [1/(Msi•in^2)]:
   2.5002e-14   2.5002e-14  -7.6802e-14
  -3.3282e-14  -3.3282e-14   1.0224e-13
  -1.7575e-14  -1.7575e-14   5.3989e-14


delta [1/(Msi•in^3)]:
   1.9106e+05  -1.1263e+05       -54187
  -1.1263e+05   1.9106e+05       -54187
       -54187       -54187   1.6646e+05


==================== Specimen 4 ====================

A [Msi•in]:
      0.36827     0.063464     0.035943
     0.063464       0.2143     0.097404
     0.035943     0.097404     0.078427


B [Msi•in^2]:
            0   6.7763e-21            0
  -6.7763e-21            0            0
            0            0            0


D [Msi•in^3]:
   3.7986e-05   1.5382e-06   5.8707e-07
   1.5382e-06   5.2917e-06   1.5909e-06
   5.8707e-07   1.5909e-06   2.5158e-06


alpha [1/(Msi•in)]:
       2.8735     -0.57958     -0.59712
     -0.57958       10.832      -13.188
     -0.59712      -13.188       29.403


beta [1/(Msi•in^2)]:
   3.6626e-17  -4.5535e-15    2.871e-15
   1.9273e-15   3.9164e-16   -6.974e-16
  -2.4104e-15   1.6005e-15  -4.4964e-16


delta [1/(Msi•in^3)]:
        26644      -7255.1      -1629.7
      -7255.1   2.3532e+05  -1.4712e+05
```

```
      -1629.7  -1.4712e+05   4.9091e+05


==================== Specimen 5 ====================

A [Msi•in]:
      0.27354     0.081206   3.6856e-19
     0.081206      0.27354   9.0598e-18
   3.6856e-19   9.0598e-18     0.096169


B [Msi•in^2]:
   -0.0016168  -6.7763e-21  -0.00026946
  -6.7763e-21   0.0016168  -0.00026946
  -0.00026946  -0.00026946            0


D [Msi•in^3]:
   2.1349e-05    1.828e-06   4.2139e-23
    1.828e-06   2.1349e-05   1.0358e-21
   4.2139e-23   1.0358e-21   2.8056e-06


alpha [1/(Msi•in)]:
       9.9306      -4.0166        2.916
      -4.0166       9.9306       -2.916
        2.916       -2.916       12.342


beta [1/(Msi•in^2)]:
       765.26       275.45       568.01
      -275.45      -765.26       568.01
        346.9        346.9   7.1351e-14


delta [1/(Msi•in^3)]:
    1.078e+05        16008        47043
        16008    1.078e+05       -47043
        47043       -47043   4.6554e+05
```

Problem 2

```
angles2 = angles - 15;
[As2,Bs2,Ds2,alphas2,betas2,deltas2] = laminate_matrices(Q, angles2, T, h);
names = {'A', 'B', 'D', 'alpha', 'beta', 'delta'};
matrices = {As2, Bs2, Ds2, alphas2, betas2, deltas2};

fprintf('\n==================== Problem 2 ====================\n');
fprintf('\n===== Sufficiently long specimen =====\n\n');

alpha11 = squeeze(alphas2(1,1,:));
E11 = 1 ./ (alpha11 * h * N_layers);
```

```
fprintf('\nEffective Tensile Modulus E11:\n');

for i = 1:N
    fprintf('Specimen %d: %.4f Msi\n', i, E11(i));
end
```

```
==================== Problem 2 ====================

===== Sufficiently long specimen =====


Effective Tensile Modulus E11:
Specimen 1: 4.0561 Msi
Specimen 2: 9.9343 Msi
Specimen 3: 3.8719 Msi
Specimen 4: 9.9343 Msi
Specimen 5: 3.4941 Msi
```

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Func 1

Func

```
function [As,Bs,Ds,alphas,betas,deltas] = laminate_matrices(Q, angles,
T_func, h)

[N, N_layers] = size(angles);

z = ((0:N_layers) - N_layers/2)*h;

As = zeros(3,3,N);
Bs = zeros(3,3,N);
Ds = zeros(3,3,N);

alphas = zeros(3,3,N);
betas  = zeros(3,3,N);
deltas = zeros(3,3,N);

% Iterates through all specimens
for i = 1:N

    A = zeros(3);
    B = zeros(3);
    D = zeros(3);

    for k = 1:N_layers
        angle = deg2rad(angles(i, k));
        transformation = T_func(angle);
        Qbar = transformation \ Q / transformation';

        z0 = z(k);
        z1 = z(k+1);

        A = A + (z1 - z0) * Qbar;
        B = B + (z1^2 - z0^2) * Qbar / 2;
        D = D + (z1^3 - z0^3) * Qbar / 3;

    end



    ABD = [A B; B' D];
    compliance = ABD \ eye(6);

    As(:,:,i) = A;
    Bs(:,:,i) = B;
    Ds(:,:,i) = D;
```

```
    alphas(:,:,i) = compliance(1:3,1:3);
    betas(:,:,i) = compliance(1:3,4:6);
    deltas(:,:,i) = compliance(4:6,4:6);

end
end
```

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